

Jacob Mattie

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Full-stack software developer with a **mathematics** background connecting data intake systems to API design and clean UI. Reliability-focused **production software** from architecture to development.

Languages

Bash
C++
Python
SQL

Data

CRUD
PostgreSQL
REST APIs
Redis

Testing

Playwright
End-to-End
Test Automation
Log Tracing

Tools

AI Harnesses
CI/CD
Docker
vim

BSc. Mathematics, 2022, Simon Fraser University

Work Experience

Quantitative Researcher — Full-Stack

River Wealth / Aligned Capital Partners

Vancouver, BC

11/2024 - 8/2026

Q: What data exists that might let us predict stock growth and market behaviour?

- Built production **data pipelines** scraping daily performance metrics for over 45,000 securities with Redis, Celery workers, and Celery-Beat scheduling within a Django framework.
- Extended a PostgreSQL database into user-facing tools through **API endpoints** and UI design.
- Designed fault-tolerant infrastructure to accommodate unreliable external data sources, surfacing errors via journalctl **log tracking** across fault tiers.
- Fully provisioned the office's **IT stack** through the configuration of a headless local linux server's systemd files through **SSH**, ensuring maximal reliability for the data engine.
- Developed bash-based **CI/CD** tooling using rsync, git, and custom logic to validate code stability when pushing changes from development environments to production.
- Automated 20%** of company-wide workload through Python scripting and process standardization.

Test Technologist

IBC Technologies

Burnaby, BC

6/2022 - 9/2023

Q: How can we safely optimize heat transfer from a boiler's combustion chamber?

- Automated hardware validation processes by leveraging IoT microcontrollers and PLC systems, streaming test data into **cloud-connected repositories** for analysis.
- Built test-validation workflows in MATLAB for multi-million-point datasets, **reducing labour time to 15%** of pre-script duration.

IT Technician (Internship)

BC Cancer Research Centre (BCCRC)

Vancouver, BC

09/2018 - 01/2019

Q: How much digital freedom can we grant researchers without compromising network security?

- Managed user access controls and permissions within a segregated research network, enforcing data isolation and **cybersecurity** policies across multiple user groups.
- Assisted in network recovery during facility-wide outage caused by a malware incident, supporting administrators in restoring connectivity and validating **system integrity** across endpoints.

Volunteering

Automation Technologist

TRIUMF - Canada's Particle Accelerator

Vancouver, British Columbia

05/2025 - Ongoing

Q: What modern technological advances have resolved longstanding engineering problems?

- Contributed the **technical foundations** of 3 scientific publications (in preparation) on distributed control systems in nuclear physics laboratories.
- Developed a containerized EPICS **deployment pipeline** (Docker + Bash), leading a modernization initiative for a legacy control software operating over \$1.5B in infrastructure.
- Coded a Raspberry Pi-based stepper motor control system, merging hardware wiring, driver design, and Python-based interface through a transistor array for **automated and repeatable** experimental control.